

COURSE 3 - DATA ENGINEERING 2026**C3-09 - Targeted Remediation Cards**

Repair the failed competency only. Preserve passing evidence; then complete a changed micro-retest.

Artifact	C3-09-REPAIR
Status	2026 - LEARNER EDITION
Phase	C3-09 Production Capstone & Engineering Transition 15 lessons 36-48 focused hours
Prerequisite	C3-08 Gate PASS; all earlier C3 evidence retained.
Assessment boundary	C3-104 is the phase Gate only. C3-113 remains the later course-wide Final Production Gate.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: requirement -> mental picture -> small proof -> guided integrated build -> expected result -> why it worked -> break/fix -> changed independent case -> validation -> production reasoning -> evidence. Integration is not mastery until the learner can explain and recover the system independently.

Repair routes

A Requirements/architecture

Rewrite the consumer outcome and measurable NFRs; compare a simpler alternative before changing tools.

B Incremental integrity

Return to stable key/version rule, replay convergence and late-correction handling.

C Transformation/serving

Restate grain at every transform; independently validate keys/values/business totals.

D Orchestration/recovery

Classify failures, bound the data interval/backfill and prove idempotent side effects.

E Spark parity

Run the same fixture through trusted reference and Spark; fix parity before tuning.

F Quality/reconciliation

Use independent keys + counts + values/totals + business invariant; fail closed.

G Observability/incident

Start from user impact, correlate evidence, contain, repair, revalidate and add prevention.

H Security/governance

Reduce broad grants; separate control/data plane; sanitize evidence; test intended identity where runtime allows.

I CI/CD

Restore test dependency/protected environment/minimum permissions and post-deploy data validation.

J Handoff/defense

Reproduce from README/runbook with a fresh reviewer; answer component/failure/evidence/scale questions without notes.